Oceanus: An Auditable LLM Governance Agent for Marine Debris Response in Aquaculture Zones

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***Abstract*—Coastal aquaculture zones face chronic floating debris—expanded-polystyrene buoy fragments, discarded fishing gear, and feed bags—that existing technology already "sees": video monitoring networks detect it and cleanup robots collect it. The bottleneck has moved from perception to decision: who decides what to clean, on what legal basis, with what oversight, and with what audit trail after the fact. We present Oceanus, a deployed-software governance agent for aquaculture-zone marine debris response that makes the decision layer itself accountable. The agent plans response actions, invokes tools, is intercepted by a policy guard on out-of-policy actions, routes them to human approval, dispatches a MQTT-connected surface robot with three-bin sorting, and replays every decision from an audit log that binds each conclusion to versioned regulatory evidence. Domain knowledge is organized as an evidence-audited ontology—100 nodes and 200 relations over 27 registered assets in the current version—in which every extracted node and relation must carry a citation that verifies verbatim against the registered source text (96% of node names and 85% of relations fully traceable). Perception runs three interchangeable channels, including an open-vocabulary zero-shot detector whose vocabulary can be extended on site without retraining; on the public SC-FloatWaste benchmark (official validation split as untouched test, 400 images, 3 regions), the zero-shot channel reaches 0.143 mAP@0.5 with precision-oriented behaviour, while a YOLO11s channel fine-tuned on 600 training images reaches 0.514 mAP@0.5 (3.6×), with cross-region generalization reported per region. All tools and knowledge are exposed to external agents through the Model Context Protocol (21 read-only tools). We report honest boundaries throughout: benchmark accuracy is not field accuracy, and the approval workflow has not yet been exercised by a real authority.**

***Index Terms*—marine debris, governance agents, LLM agents, auditability, human-in-the-loop, open-vocabulary detection, Model Context Protocol**

I. Introduction

Aquaculture zones along the Chinese coast accumulate floating debris at a rate that makes manual patrols uneconomical: expanded-polystyrene (EPS) buoy fragments from rafts, derelict fishing gear, and feed bags. The technology landscape already covers the two ends of the problem. Video monitoring networks—deployed at city and county scale since at least 2020—detect and quantify debris. Cleanup platforms such as WasteShark and Clearbot collect it, one machine at a time. What is missing is the layer between: a decision process that turns detections into dispatched, legally grounded, humanly approved cleanup actions—and that can be audited afterwards.

This gap is not a perception problem but an accountability problem. A governance decision ("dispatch the robot to grid cell X because debris type Y exceeds threshold Z under regulation W") must cite its basis, survive challenge, and be replayable. Large language model (LLM) agents are now capable of the planning and tool-use parts, but exactly their fluency makes auditability non-negotiable: an agent that explains itself convincingly is not the same as an agent whose every claim traces to registered evidence.

Oceanus is a software system built around this observation. Its contribution is not a better detector but a decision loop with accountability as a design constraint:

1) **An auditable decision pipeline** (Sect. IV): planning → tool invocation → policy guard → human approval → dispatch → audit replay, with every decision bound to versioned knowledge assets, an ontology version, and a policy version.
2) **Evidence-audited domain knowledge** (Sect. VI): a versioned ontology in which every node and relation extracted by rules or an LLM must cite a passage that verifies verbatim against registered source documents; a deterministic audit reports traceability rates.
3) **Interchangeable perception, honestly measured** (Sect. V): three channels—a temporal OpenCV channel for production, an open-vocabulary zero-shot channel whose vocabulary extends on site without retraining or restart, and a supervised channel—evaluated on a public benchmark with region-level reporting and negative results reported as such.
4) **Standard-protocol interoperability** (Sect. VI): knowledge, retrieval, events, and work orders are exposed as 21 read-only tools over the Model Context Protocol (MCP), so external agents can use the same evidence-backed capabilities.

II. Related Work

Marine debris and monitoring. Debris harms marine life across taxa [1], and computer-vision detection of floating and river plastic is an active field in its own right [2]. Coastal governments have deployed video monitoring and cleanup programmes for years; public tenders and municipal systems at city scale have existed in China since 2020. Commercial cleanup platforms—WasteShark [3], Clearbot [4]—are single-machine products with onboard collection. Both strands treat decision-making as a human task. Oceanus keeps both ends replaceable— cameras and robots are endpoints—and concentrates the contribution in the decision layer. Compared on price

TABLE I
QUALITATIVE COMPARISON WITH CLEANUP PLATFORMS. WASTESHARK LIST PRICE FROM PUBLIC LISTING (APPROX. USD 23,600), CONVERTED AT 7.2 CNY/USD.

	Cleanup platforms	Oceanus
Form factor	single machine	auditable agent + replaceable endpoints
Decision	manual	plan → guard → approve → execute → replay
New classes	product change	add a vocabulary word on site
Basis	none	versioned regulatory evidence chain
Sorting	none / single bay	three-bin coarse sorting + shore-side fine sorting
Indicative cost	∼USD 23.6k (WasteShark)	commodity hardware, open stack

and capability (Table I), single-machine products leave the governance workflow unbuilt.

LLM agents, tool use, and auditability. LLM-based autonomous agents [5] couple planners to tool APIs; retrieval-augmented generation [6] grounds answers in retrieved text. Our concern is not planner design but the guardrails around it: deterministic policy checks before execution, human approval for out-of-policy actions— a human-in-the-loop pattern with long precedent in interactive machine learning [7]—and replayable traces that bind each step to versioned knowledge.
Open-vocabulary detection. YOLO-World [8] detects arbitrary text prompts zero-shot; supervised detectors such as YOLO11 [9] lead in accuracy when training data exists. We use the former as an operational capability (inspectors name a new debris class on site) and measure its real accuracy profile on a public benchmark [10] rather than assuming it.

III. SYSTEM ARCHITECTURE

Oceanus organizes a three-tier loop: (i) *perception*—shore-based cameras with an edge box running detection; (ii) *decision*—the governance agent on the data platform; (iii) *execution*—a surface robot connected over MQTT, with three-bin coarse sorting on board and shore-side fine sorting. Data flows back from operations into heat maps, work orders, and reports, closing the governance loop. The software comprises five subsystems: edge sensing, onboard robot software, the data platform (FastAPI, PostgreSQL with PostGIS, Redis, EMQX MQTT, MinIO, go2rtc streaming, Vue 3 front end), a training pipeline, and an inference service. The robot is a replaceable MQTT work endpoint: with it offline, the platform still completes event understanding, planning, guard checks, approval, dispatch, audit, and replay—the configuration under which the system currently runs.

A. Execution endpoint protocol

Execution endpoints (edge boxes, robots) attach over MQTT through a fixed topic tree, designed for the shore-side reality of intermittent 4G coverage and solar-powered devices: retained status topics with last-will messages give immediate offline detection, and the protocol deliberately avoids QoS 2—deduplication is enforced at the business layer with a `(device_id, seq)` unique constraint, since protocol-level exactly-once delivery does not remove the need for idempotent ingestion. Telemetry (heartbeats, battery, bin fill levels) is sent QoS 0 and may drop frames; events and commands are QoS 1. Site identity lives in the topic hierarchy (`marine/{site}/{device}/...`), making the site the access-control boundary without per-device scripting. Work orders follow an explicit state machine, `pending` → `assigned` → `navigating` → `collecting` → `done` (with cancellation from any live state), and an unacknowledged assignment rolls back to `pending` and is redispatched—the robot is thus a replaceable endpoint whose failure modes are protocol-visible rather than silent.

TABLE II
MQTT TOPIC CONTRACT (SUBSET). ALL TOPICS ARE DEVICE-TO-PLATFORM EXCEPT `CMD` AND `ROBOT/{ID}/...`, WHICH ARE PLATFORM-TO-DEVICE.

Topic	QoS	Retain
`marine/{site}/{dev}/event`	1	no
`marine/{site}/{dev}/telemetry`	0	no
`marine/{site}/{dev}/status`	1	LWT
`marine/{site}/{dev}/cmd`	1	no
`marine/{site}/{dev}/cmd/ack`	1	no
`robot/{id}/task`	1	no
`robot/{id}/task/progress`	1	no
`robot/{id}/cmd/ack`	1	no

LWT: retained last-will message (immediate offline detection).

IV. THE AUDITABLE DECISION PIPELINE

The agent runtime executes a fixed accountability envelope around every response action (Fig. 2):

1) **Planning.** Given an event (detection cluster, location, time), the agent plans a response using registered tools.
2) **Policy guard.** Before execution, planned actions pass a deterministic policy check (scope, budget, zone restrictions, action-type rules). Out-of-policy plans are blocked, not executed.
3) **Human approval.** Blocked or high-impact actions route to an approval queue; nothing executes until a human approves.
4) **Dispatch and execution.** Approved actions dispatch work orders (MQTT to the robot when online; queued otherwise).
5) **Audit replay.** Every step—plan, guard verdict, approval, dispatch, outcome—is logged with the knowledge-asset versions, ontology version, and policy version that were live at decision time, so any past decision can be reconstructed exactly.

The knowledge-backed question-answering layer follows the same discipline: retrieval first, then generation with inline numbered citations; when evidence is insufficient the system

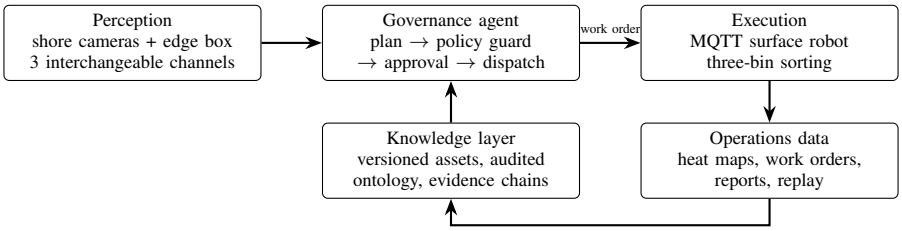


Fig. 1. Three-tier governance loop. The knowledge layer binds every agent decision to versioned evidence; operations data flows back into knowledge and reporting.

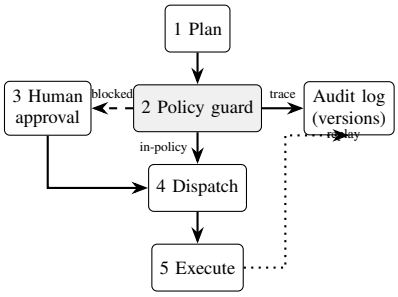


Fig. 2. The accountability envelope around every response action. The policy guard blocks out-of-policy plans before execution; blocked or high-impact actions require human approval; every step is logged with knowledge-asset, ontology, and policy versions for later replay.

states a refusal instead of generating, and if the LLM call fails it degrades to presenting the raw evidence list. Refusal is decided on *evidence coverage*, not retrieval score; we later show why (score-based abstention is configuration-fragile on small regulatory corpora).

V. Perception Channels and Public-Benchmark Evaluation

A. Channels

Three interchangeable channels share one interface: (i) a production OpenCV channel (background modeling plus color/shape rules) that depends on continuous frames, runs on CPU, and is tuned per site; (ii) an *open-vocabulary* zero-shot channel (YOLO-World) whose vocabulary is a runtime word list—an inspector can name "foam buoy fragment" on site and the detector picks it up without retraining or restart; and (iii) a supervised YOLO11s channel for accuracy when training data exists. The channels are compared side by side in the product UI under a `/vision-compare` view so their operating regimes are visible rather than claimed.

B. Evaluation protocol

Static-image accuracy is measured on the public SC-FloatWaste benchmark (CC BY 4.0; 7,906 images, 27,583 boxes, 25 fine classes across three regions: Indonesia/Citarum, Australia NSW, Australia Tasmania). We use the *official* train/val split, which the dataset publisher declares disjoint: the official validation split (400 images, 1,490 boxes) is our untouched test set; 600 official-train images from Indonesia form the training set and 172 a validation set. Manifest selection is deterministic (within-region filename order; no random splits, which would leak adjacent video frames). Model selection uses only the validation split (best validation mAP@0.5 at epoch 48); the test set was never used for any tuning choice. The zero-shot vocabulary was frozen in code *before* the benchmark was adopted (plastic bottle, plastic waste, garbage, fishing net, foam debris, marine debris, discarded shoe). Every metric JSON records commands, dates, weight hashes, manifests, and claim boundaries.

C. Findings, including negative ones

Table III and Table IV support four findings. (1) Supervised training pays: fine-tuned YOLO11s lifts mAP@0.5 from 0.143 to 0.514 (3.6×; best F1 0.5376 at precision 0.5665 / recall 0.5114). (2) The zero-shot channel is precision-oriented: at confidence 0.05 it holds precision 0.5054 at recall 0.0631, and its dominant failure mode is misses, not false alarms (Indonesia: 589 detections against 850 ground-truth boxes). This is the accuracy profile of a *supplementary on-site extension channel*, not a primary alarm channel—which is exactly how the product deploys it. (3) Raising input resolution *lowered* zero-shot AP (0.143→0.116): higher resolution produced more low-confidence false positives, so operating points must be calibrated per channel. (4) Longer training traded cross-domain generalization for in-domain accuracy: an early checkpoint scored 0.443 on NSW while the fully-trained model scores 0.334 there (in-domain rose 0.627→0.723). Domain adaptation needs multi-region data, not more epochs—one reason local data collection is prioritized over public-benchmark tuning.

D. Honest boundaries

These are third-party public-benchmark numbers on river and urban waterway imagery. They are *not* field validation in the target aquaculture zones, whose EPS-dominant debris, sea-state, and lighting distributions differ; the local test set is still empty pending a collection campaign, and no on-site accuracy claim is made. The production OpenCV channel requires continuous frames and is not evaluable on static-image benchmarks; its 76% figure in project materials is a temporal false-alarm suppression rate on a different pipeline and must not be mixed with the benchmark numbers.

TABLE III
MAIN RESULTS ON THE UNTOUCHED TEST SPLIT (400 IMAGES, 1,490 BOXES, 3 REGIONS, SINGLE CLASS "LITTER"). AP COMPUTED WITH ALL-POINT INTERPOLATION AT MODEL-SIDE CONFIDENCE FLOOR 0.001 (NOT THE OPERATING THRESHOLD).

Channel	mAP@0.5	mAP@50:95	best F1
Zero-shot (YOLO-World [8], imgsz 960)	0.1432	0.0930	0.2702
Zero-shot (YOLO-World, imgsz 1280)	0.1155	0.0744	0.2417
Fine-tuned (YOLO11s [9], imgsz 1280)	**0.5135**	**0.3444**	**0.5376**

TABLE IV
PER-REGION RESULTS (MAP@0.5). TRAINING USED INDONESIA ONLY; NSW AND TASMANIA ARE FULLY CROSS-DOMAIN.

Region (images / boxes)	Zero-shot	Fine-tuned	Domain
Indonesia (90 / 850)	0.1299	**0.7229**	training
NSW (164 / 204)	0.2105	**0.3341**	cross
Tasmania (146 / 436)	0.2408	**0.4029**	cross
All (400 / 1,490)	0.1432	**0.5135**	—

VI. KNOWLEDGE ASSETS AND INTEROPERABILITY

A. Versioned assets and evidence-audited ontology

Documents, tables, images, events, telemetry, and datasets register as immutable-versioned assets. A domain ontology (current published version: 100 nodes, 200 relations over 27 assets, primarily marine dumping regulations from the national environmental authority) is built through candidate extraction $\rightarrow$ human review $\rightarrow$ publish. The auditable-by-construction invariant: every node and relation carries at least one evidence citation $\langle$asset, version, snippet$\rangle$. A deterministic audit re-verifies each snippet verbatim against the registered text and checks endpoint occurrence in citing documents: currently 89.9% of citations verify verbatim (302/336), 90% of relation endpoints are traceable, 85% of relations are fully traceable, and 96% of node names occur in their source document. Non-verbatim citations feed the review queue rather than being silently accepted. Retrieval over these assets is multi-hop (ontology graph expansion) and every decision stores its evidence chain—asset versions, ontology version, policy version—for replay. On a 306-question type-stratified benchmark over this corpus (title, entity, two-hop, and refusal queries, deterministically constructed from the ontology and asset tables), the deployed graph-expanded retriever reaches 98.3% hit@5 on two-hop queries and 93.0% on entity queries, and a two-system rule router that sends title-coverage queries to plain BM25 and everything else to the graph retriever attains 97.0% overall hit@5, statistically matching a per-query-type oracle—while no configuration's retrieval scores separate answerable from out-of-domain queries, which is why the QA layer abstains on evidence coverage rather than score.

The question-answering workbench enforces the same discipline at the user interface: answers carry inline numbered citations whose badges locate the grounding evidence card (including hop paths and asset-version references); when retrieved evidence is insufficient the interface shows an explicit refusal banner instead of a generated answer, and if the LLM call itself fails the system degrades to presenting the raw evidence list. Refusal is thus a first-class output, not an error path.

B. Deployment status

The decision-loop software and the MCP service run today as a single-node cloud deployment: a FastAPI backend with PostgreSQL/PostGIS, Redis, EMQX and MinIO, fronted by the MCP server (stdio and Streamable HTTP). The 21-tool read-only surface has been exercised end-to-end by an external hosted agent platform: real question-answering calls through knowledge search and event tools completed successfully, with server-side logs matching the platform's call traces step by step. The robot remains a replaceable MQTT endpoint (offline in the current configuration); no field campaign has yet been run, so no on-site accuracy or cleanup-effectiveness claim is made anywhere in this paper.

C. MCP interoperability

The same capabilities are exposed to external agents through the Model Context Protocol [11]: 21 read-only tools (Table V) over stdio or Streamable HTTP, with write tools disabled unless explicitly enabled. Outbound calls authenticate with a dedicated service account and refresh tokens before expiry; inbound HTTP uses a bearer token. Five reusable workflow skills ship with the server. Any MCP-capable agent can therefore pose questions and receive citation-backed answers from the same versioned evidence the governance decisions use—interoperability without giving up the audit chain.

VII. LIMITATIONS

(1) No field deployment validation yet: perception accuracy is public- benchmark only, and the aquaculture-zone test set is pending. (2) The approval workflow has not been exercised by a real authority; the human role is currently project staff. (3) The robot end is a replaceable MQTT endpoint; end-to-end loops including physical collection have not been validated at sea. (4) The governance domain is one Chinese coastal county's regulatory environment; the ontology and policy sets are locale-specific by design. (5) Open-vocabulary detection's low recall on cross-domain imagery limits its role to a supplementary channel.

TABLE V
THE 21 READ-ONLY MCP TOOLS BY CATEGORY (WRITE TOOLS EXIST BUT ARE DISABLED UNLESS EXPLICITLY ENABLED).

Category (n)	Tools
Knowledge (7)	`knowledge_` prefix: `list_assets`, `asset_detail`, `list_ontology_versions`, `ontology_version_detail`, `list_ontology_nodes`, `list_ontology_relations`, `search`
Decision evidence (2)	`knowledge_list_decisions`, `knowledge_decision_evidence`
Events (2)	`event_list`, `event_get`
Work orders (3)	`task_list`, `task_get`, `task_ack_history`
Dashboard (1)	`dashboard_get`
Agent runtime (6)	`agent_runtime_status`, `agent_list_runs`, `agent_run_detail`, `agent_run_steps`, `agent_list_tools`, `agent_list_approvals`

VIII. CONCLUSION

Oceanus moves marine debris response from "can we see it?" to "can we defend the decision?". Its decision loop—plan, guard, approve, execute, replay—binds every action to versioned, verbatim-verifiable evidence; its knowledge layer is audited deterministically rather than trusted; its perception is honestly measured on a public benchmark with negative results reported; and everything is reachable by external agents over an open protocol. The system runs today as decision-loop software with replaceable perception and execution endpoints; the next step is the aquaculture-zone field campaign that will turn public-benchmark accuracy into local accuracy, under the same audit discipline.

DATA AVAILABILITY

The public-benchmark evaluation artifacts (metric JSONs with commands, weight hashes, manifests, and claim boundaries), the retrieval benchmark, and all evaluation scripts are available in the project repository. The SC-FloatWaste dataset is publicly available under CC BY 4.0 (doi: 10.5281/zenodo.21934286); the regulatory corpus consists of publicly published government documents.

REFERENCES

[1] S. C. Gall and R. C. Thompson, "The impact of debris on marine life," *Marine Pollution Bulletin*, vol. 92, no. 1–2, pp. 170–179, 2015.

[2] N. Maharjan, H. Miyazaki, B. M. Pati, M. N. Dailey, and colleagues, "Detection of river plastic using UAV sensor data and deep learning," *Remote Sensing*, vol. 14, no. 13, art. 3049, 2022.

[3] RanMarine Technology, "WasteShark aquadrone." [Online]. Available: https://www.ranmarine.io

[4] Clearbot, "AI-powered autonomous cleaning boats." [Online]. Available: https://www.clearbot.org

[5] L. Wang, C. Ma, X. Feng, Z. Zhang, H. Yang, J. Zhang, Z. Chen, J. Tang, X. Chen, Y. Lin, W. X. Zhao, Z. Wei, and J.-R. Wen, "A survey on large language model based autonomous agents," *Frontiers of Computer Science*, vol. 18, no. 6, 2024.

[6] P. Lewis, E. Perez, A. Piktus *et al.*, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Proc. NeurIPS*, 2020.

[7] X. Wu, L. Xiao, Y. Wu, Q. Z. Sheng, and S. Wang, "A survey on human-in-the-loop machine learning," *ACM Computing Surveys*, vol. 55, no. 13s, pp. 1–35, 2022.

[8] Y. Xu, T. Zhang, M. Zhang, R. Guo, G. Tian, Y. Zhang, X. Hu, X. Wang, and J. Dai, "YOLO-World: Real-time open-vocabulary object detection," in *Proc. IEEE/CVF CVPR*, 2024.

[9] Ultralytics, "YOLO11 object detection model," 2024. [Online]. Available: https://github.com/ultralytics/ultralytics

[10] O. Wu, "SC-FloatWaste: A multi-region floating waste detection dataset," Zenodo, CC BY 4.0. doi: 10.5281/zenodo.21934286.

[11] Anthropic, "Model Context Protocol," 2024. [Online]. Available: https://modelcontextprotocol.io